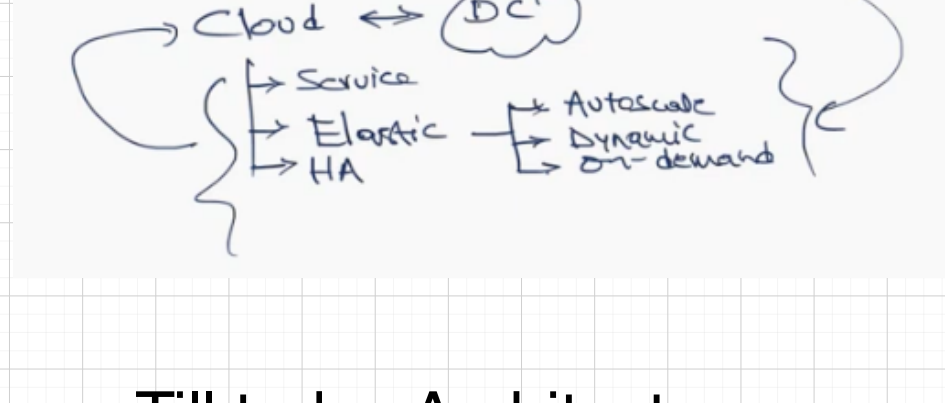
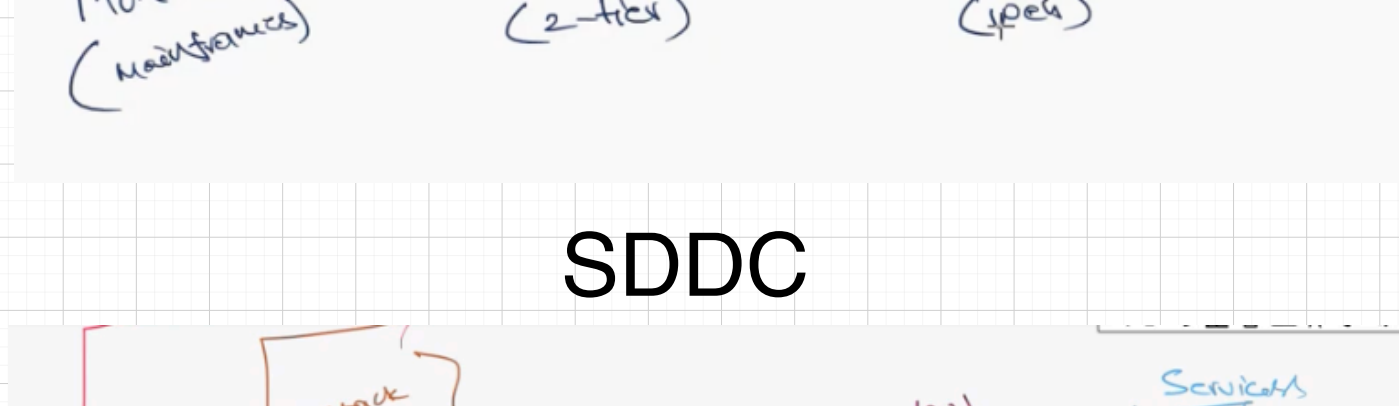


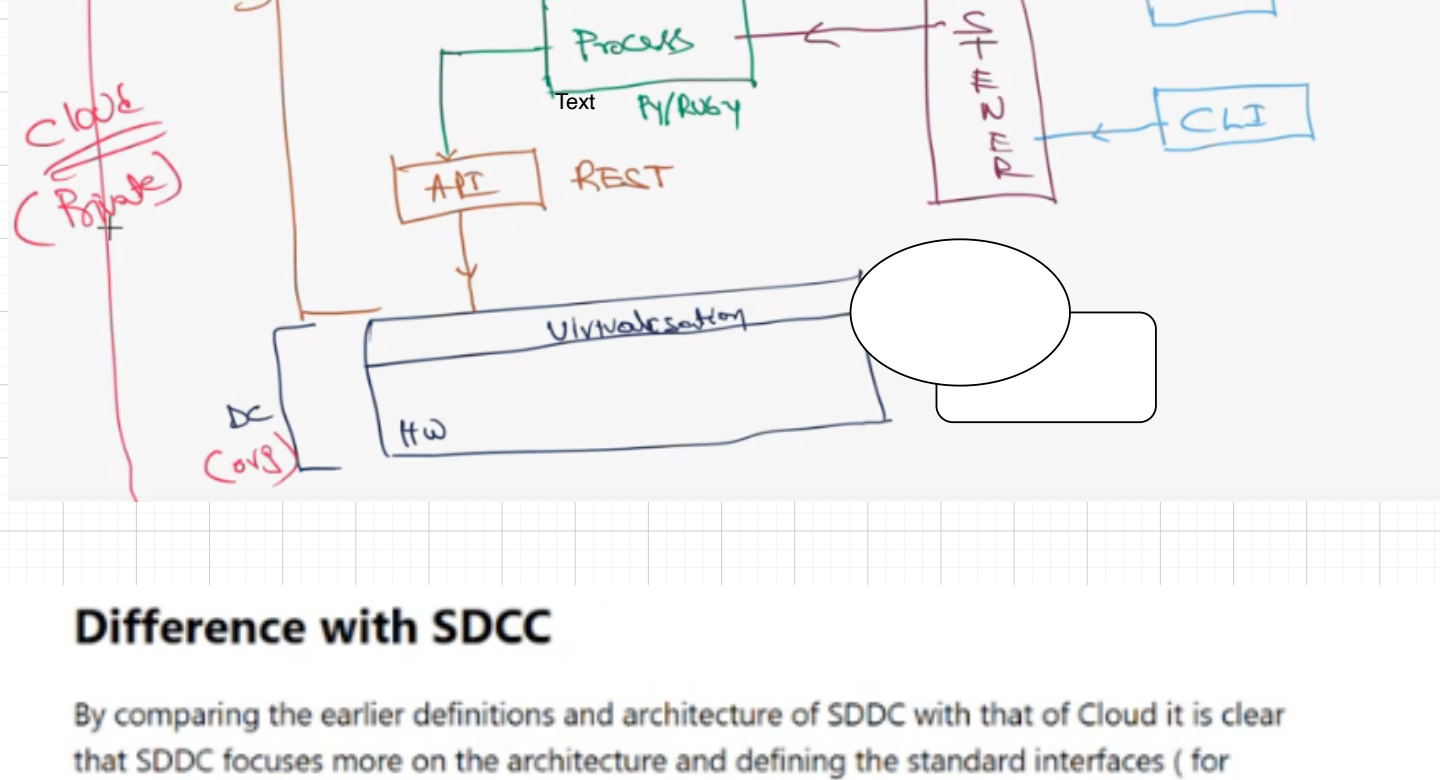
What is a Cloud



Till today Architecture



SDDC



Difference with SDCC

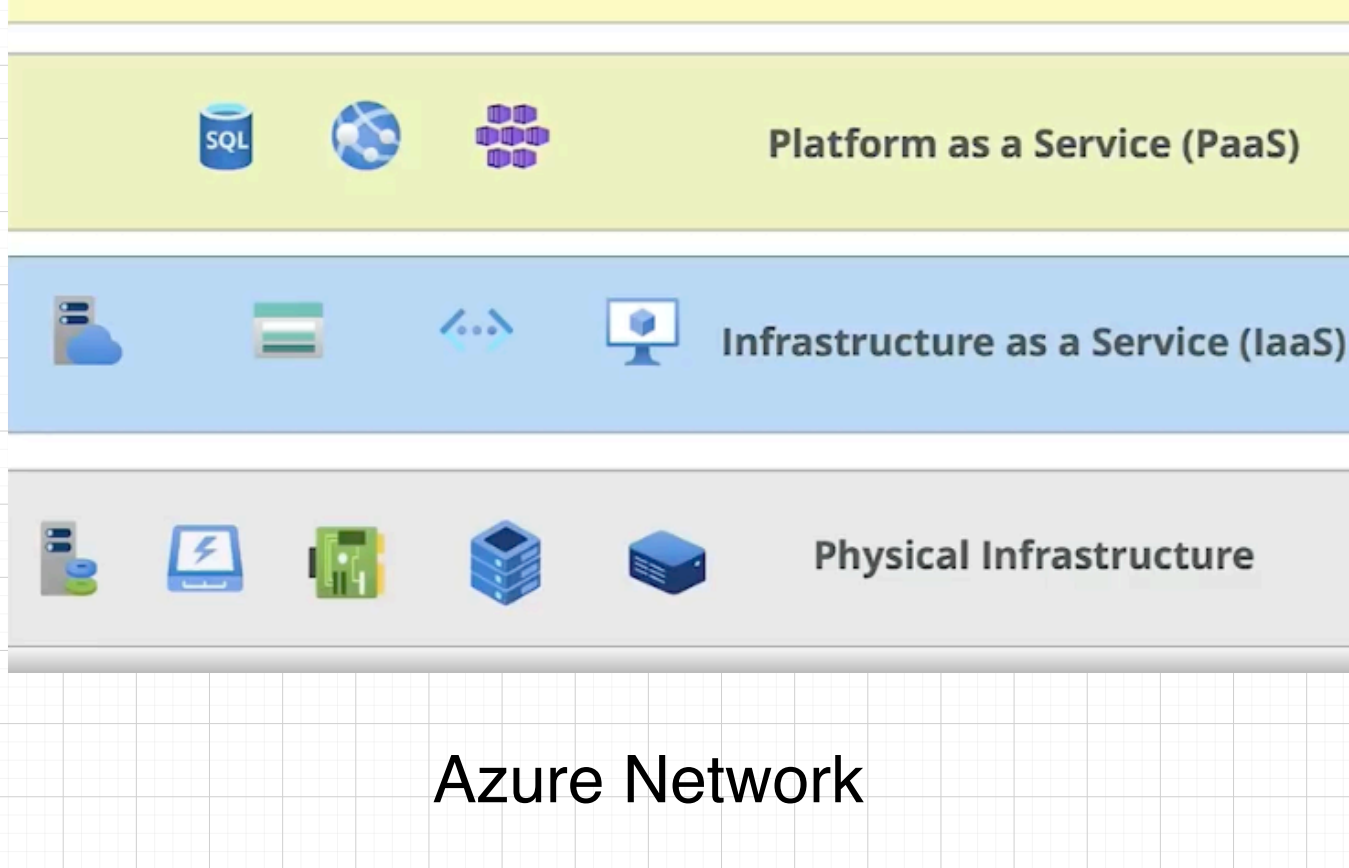
By comparing the earlier definitions and architecture of SDDC with that of Cloud it is clear that SDDC focuses more on the architecture and defining the standard interfaces (for example datacenter abstraction layer) while cloud is more focused on the services and capabilities. However, if you read the definition of cloud clearly, there is nothing in this definition that the SDDC is not able to provide.

Therefore we can say that "Software-defined data center (SDDC) provides the components and architecture to build a cloud. Further, Open SDDC is one way to build the cloud. But there could be other ways to build the cloud, for example, any vendor-specific architecture.

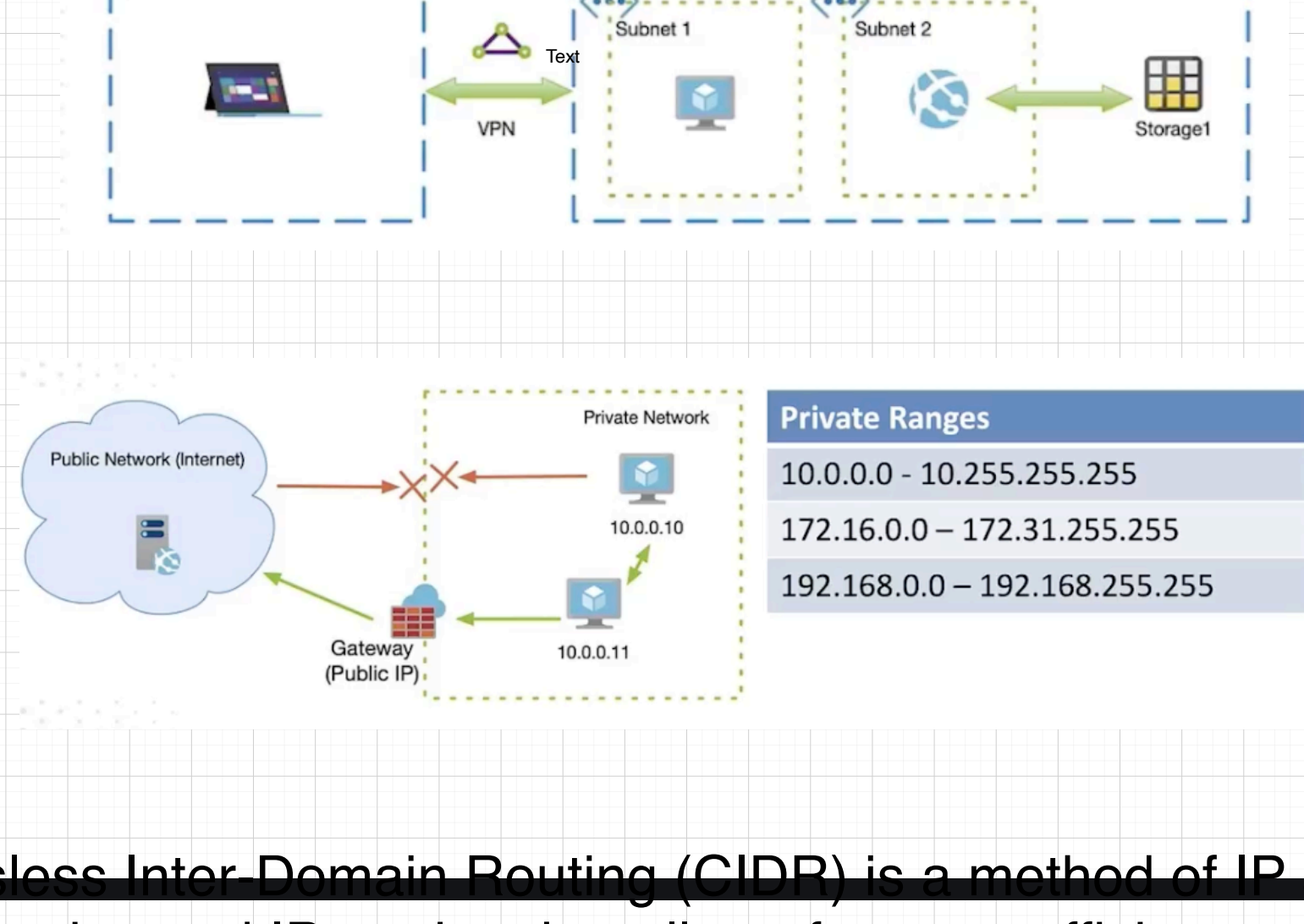
To make it simple we use "SDDC" to build "cloud"

Its your turn to tell me what is your understanding of SDDC and cloud? and do you agree with this way of explanation?

Cloud Services



Azure Network

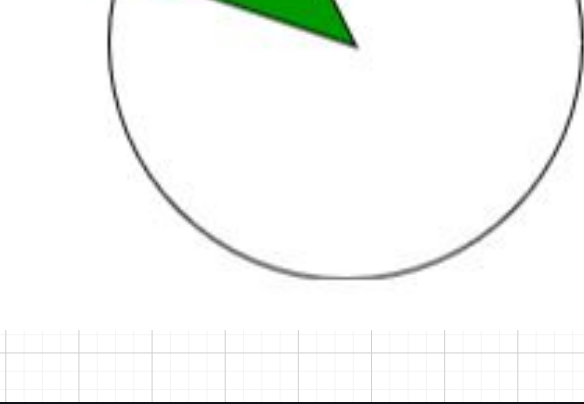


Classless Inter-Domain Routing (CIDR) is a method of IP address allocation and IP routing that allows for more efficient use of IP addresses. CIDR is based on the idea that IP addresses can be allocated and routed based on their network prefix rather than their class, which was the traditional way of IP address allocation.

In Classful addressing the no of Hosts within a network always remains the same depending upon the class of the Network.

Class A network contains 2^{24} (IP addresses) or $2^{24} - 2$ Hosts,
Class B network contains 2^{16} (IP addresses) or $2^{16} - 2$ Hosts,
Class C network contains 2^8 (IP addresses) or $2^8 - 2$ Hosts

Now, let's suppose an Organization requires 2^{14} hosts, then it must have to purchase a Class B network. In this case, 49150 Hosts will be wasted. This is the major drawback of Classful Addressing. In order to reduce the wastage of IP addresses a new concept of **Classless Inter-Domain Routing** is introduced. Nowadays IANA is using this technique to provide IP addresses. Whenever any user asks for IP addresses, IANA is going to assign that many IP addresses to the User.



Representation: It is as also a 32-bit address, which includes a special number that represents the number of bits that are present in the Block Id.

a . b . c . d / n

Where n is the number of bits that are present in Block Id / Network Id.

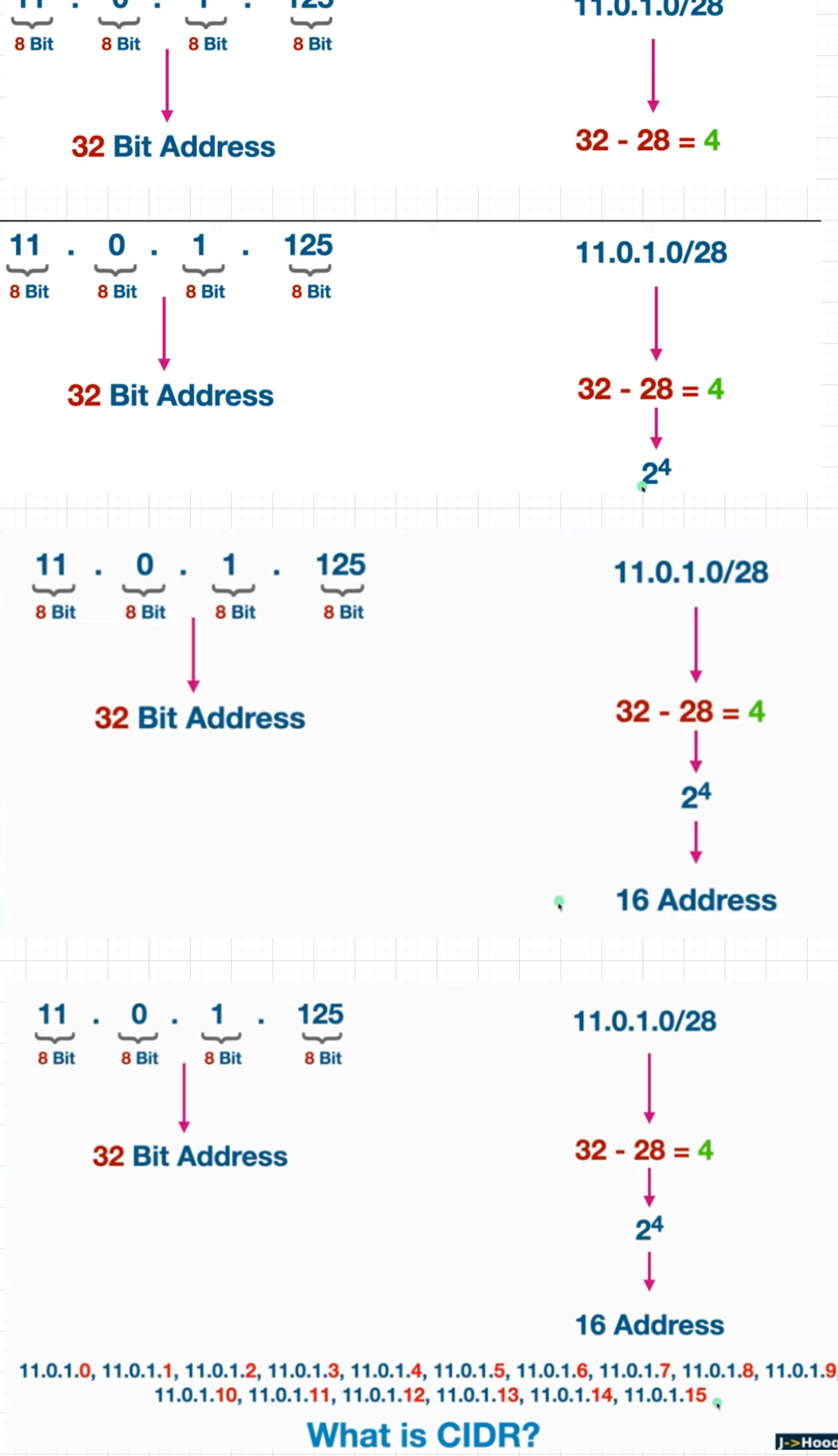
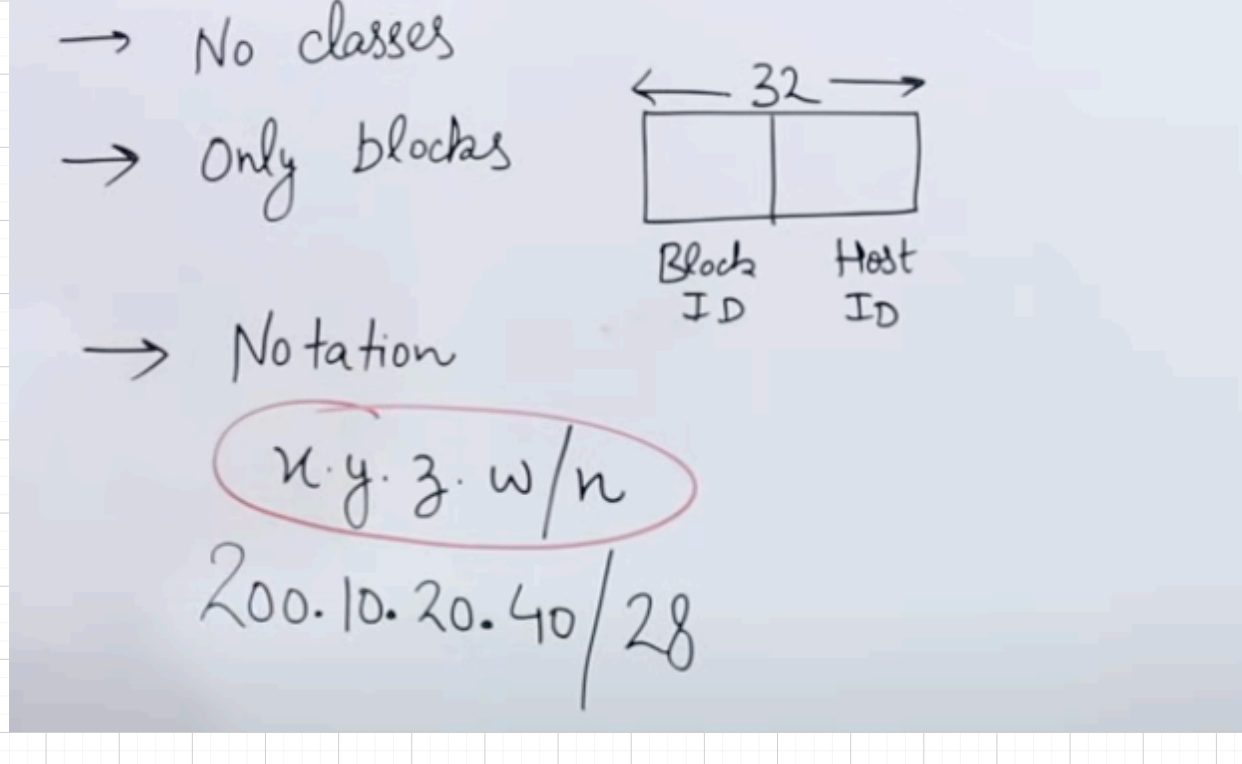
Example:

20.10.50.100/20

Rules for forming CIDR Blocks:

All IP addresses must be contiguous.

Block size must be the power of 2 (2^n). If the size of the block is the power of 2, then it will be easy to divide the Network. Finding out the Block Id is very easy if the block size is of the power of 2.



What is CIDR?

HOOD

Understanding CIDR – Subnet Mask

The Subnet Mask basically allows part of the underlying IP to get additional next values from the base IP

192.168.0.0 /32 => allows for 1 IP (2^0)	192.168.0.0 -> 192.168.0.1
192.168.0.0 /31 => allows for 2 IP (2^1)	192.168.0.0 -> 192.168.0.3
192.168.0.0 /30 => allows for 4 IP (2^2)	192.168.0.0 -> 192.168.0.7
192.168.0.0 /29 => allows for 8 IP (2^3)	192.168.0.0 -> 192.168.0.15
192.168.0.0 /28 => allows for 16 IP (2^4)	192.168.0.0 -> 192.168.0.31
192.168.0.0 /27 => allows for 32 IP (2^5)	192.168.0.0 -> 192.168.0.63
192.168.0.0 /26 => allows for 64 IP (2^6)	192.168.0.0 -> 192.168.0.127
192.168.0.0 /25 => allows for 128 IP (2^7)	192.168.0.0 -> 192.168.0.255
192.168.0.0 /24 => allows for 256 IP (2^8)	192.168.0.0 -> 192.168.0.255
...	...
192.168.0.0 /16 => allows for 65,536 IP (2^{16})	192.168.0.0 -> 192.168.255.255
...	...
192.168.0.0 /0 => allows for All IPs	0.0.0.0 -> 255.255.255.255

Quick Memo			
Octets			
1 st	2 nd	3 rd	4 th
• /32 – no octet can change			
• /24 – last octet can change			
• /16 – last 3 octets can change			
• /8 – last 3 octets can change			
• /0 – all octets can change			

